

CAC Barcode Guide for Kids Who Can't Read Good and Who Wanna Learn to Do Other Stuff Good Too

A from-scratch guide to pulling **EDIPI**, **Personnel Category Code**, and **Branch Code** out of the 18-character barcode on the back of a U.S. Department of Defense Common Access Card (CAC).

The two source specs:

- **CAC End-Point Implementation Guide v2.1.1** (2 April 2010) — high-level reference. [https://www.cac.mil/Portals/53/Documents/CAC_End_Point_Implementation_Guide_v2.1.1\(2010.4.2\).pdf](https://www.cac.mil/Portals/53/Documents/CAC_End_Point_Implementation_Guide_v2.1.1(2010.4.2).pdf)
- **DoD ID Bar Code Formats — SDK v7.5.0** (September 2012) — contains the field layout (§2.1, Table 2), the base-32 algorithm (Appendix B.1), and the lookup tables (Tables 25–27 in Appendix C). Section and table numbers below refer to this document. <https://archive.org/details/DoDIDBarCodeSDKFormatsV750Sep2012>

The CAC carries machine-readable data in three places: the embedded chip, a PDF417 barcode on the **front** (large, 2D), and a Code 39 barcode on the **back** (small, linear). **This document covers only the Code 39 barcode on the back.** Your scanner does the Code 39 decoding itself and hands your program an ordinary 18-character ASCII string; your only job is to interpret it.

From **SDK v7.5.0, §2.1, Table 2 (pp. 11–12)**, the barcode contains seven fixed-width fields packed into 18 characters with **no separators**:

Position (1-indexed)	Field	Length	What it is
1	Bar Code Version Code (VC)	1	Always "1" for this layout.
2–7	Person Designator Identifier (PDI)	6	Base-32-encoded 9-digit ID. Historically the SSN; on modern cards a synthetic card-identifier starting with 999 (see §4).
8	Person Designator Type Code (PDT)	1	Letter saying what kind of ID the PDI is (Table 25).
9–15	DoD EDI Person Identifier (EDIPI)	7	Base-32-encoded 10-digit DEERS identifier — the durable, unique person ID.
16	Personnel Category Code (PCC)	1	Letter saying what category of person the cardholder is (Table 27).
17	Branch Code (BC)	1	Letter or digit identifying branch of service (Table 26).
18	Card Instance Identifier (CI)	1	Random character distinguishing re-issued cards. Ignore for decoding purposes.

- Positions are 1-indexed in the spec; in 0-indexed Python those become `d[0]`, `d[1:7]`, `d[7]`, `d[8:15]`, `d[15]`, `d[16]`, `d[17]`.
- Length is always exactly 18. Anything else is not a CAC Code 39 barcode (or the scanner is misconfigured).
- All letters are uppercase — normalise with `.upper()` to be safe.

For just EDIPI + PCC + Branch you only need positions 9–15, 16, and 17.

A 10-digit decimal EDIPI would eat 10 characters of an 18-character budget. The spec instead writes the EDIPI (and PDI) in **base 32**: 10 decimal digits fit in 7 base-32 digits, 9 fit in 6.

2.1 The alphabet

Value:	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Symbol:	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V

Digits 0 – 9 are themselves; letters A – V are 10–31. The letters W, X, Y, Z are **not used**.

SDK v7.5.0, p. 42: "thirty-two symbols represented by the ten decimal digits 0 through 9 and the first twenty-two alphabetic characters A through V... A represents the 11th symbol... V represents the 32nd."

This is the standard "extended-hex" base-32 alphabet — and exactly the alphabet Python's built-in `int(s, 32)` already understands.

2.2 In one line of Python

```
edipi = int(barcode[8:15], 32)      # base-32 → integer
edipi_str = f"{edipi:010d}"         # zero-pad to 10 digits
```

The zero-pad matters: without it a short EDIPI like 0123456789 would print as 123456789. Same idea for the PDI: pad to 9 digits.

Per Table 2 footnotes (p. 12): the 9-digit SSN range 000-00-0000 to 999-99-9999 maps to base-32 000000 – TPLIFV ; the 10-digit EDIPI range 1000000000 – 9999999999 maps to base-32 0TPLIG0 – 9A0NOVV .

If you want the math by hand: for digits d₁ d₂ ... d_n (left to right), value = d₁·32⁽ⁿ⁻¹⁾ + d₂·32⁽ⁿ⁻²⁾ + ... + d_n·32⁰. The spec's worked example (p. 42) converts V62G to (31·32³)+(6·32²)+(2·32)+16 = 1,022,032.

3.1 Branch (Service) Codes — Table 26, p. 50

Position 17 of the barcode.

Code	Branch / Service
A	USA (Army)
C	USCG (Coast Guard)
D	DoD
F	USAF (Air Force)
H	USPHS (Public Health Service)
M	USMC (Marine Corps)
N	USN (Navy)
O	NOAA
1	Foreign Army
2	Foreign Navy

Code	Branch / Service
3	Foreign Marine Corps
4	Foreign Air Force
X	Other / Not Applicable

3.2 Personnel Category Codes — Table 27, p. 51

Position 16 of the barcode.

Code	Category
A	Active duty member
B	Presidential Appointee
C	DoD civil service employee
D	100% disabled American veteran
E	DoD contract employee
F	Former member (20-yr active-duty, eligible to retire, chose discharge)
G	National Guard, mobilised or on active duty 31+ days
H	Medal of Honor recipient
I	Non-DoD civil service employee
J	Academy student (excludes OCS)
K	Non-appropriated fund (NAF) DoD employee
L	Lighthouse service
M	Non-Government agency personnel (e.g. American Red Cross)
N	National Guard, not on active duty or active 30 days or less
O	Non-DoD contract employee
Q	Reserve retiree not yet eligible for retired pay
R	Retired Uniformed Service member eligible for retired pay
S	Reserve, mobilised or on active duty 31+ days
T	Foreign military member
U	Foreign national employee (DoD or non-DoD)
V	Reserve, not on active duty or active 30 days or less
W	DoD beneficiary (e.g. former or surviving spouse)
Y	Retired DoD Civil Service Employee

Letters not listed (P , X , Z , ...) are undefined and should fall through to "unknown". P appears in the older **Member** Category Code (TAMP) but not in the modern Personnel Category Code.

Table 25 (PDT codes — S =SSN, N / P / D / F / T / I for various non-SSN identifiers) is in the SDK doc but is not needed for the three fields above.

Originally the PDI held the cardholder's real nine-digit SSN in base-32. The DoD has since stopped putting real SSNs on cards; the PDI now contains a synthetic card-identifier whose leading three digits are always **999** (an invalid SSN range), with the remaining six digits giving 10^6 unique values per person. The PDT field will still typically be **S** for legacy compatibility.

The DoD "Appendix C — Sample Barcodes" document gives:

```
1T0QQG0S14PC0MIAA5
```

Splitting by the field widths from §1:

Positions	Slice	Decoded
1	1	Version code (must be 1).
2–7	T0QQG0	PDI = 999123456 (synthetic; see §4).
8	S	PDT = SSN.
9–15	14PC0MI	EDIPI = 1234567890 .
16	A	Personnel Category = Active duty member .
17	A	Branch = USA (Army) .
18	5	Card instance (ignore).

Verify in Python: `int("14PC0MI", 32) → 1234567890`, `f"{1234567890:010d}" → "1234567890"`.

(Beware: in scanned/OCR copies of the spec the digit **0** is often misread as the letter **O**. A real scanner hands you the correct characters.)

1. **CAC End-Point Implementation Guide v2.1.1**, DMDC, 2 April 2010. [https://www.cac.mil/Portals/53/Documents/CAC_End_Point_Implementation_Guide_v2.1.1\(2010.4.2\).pdf](https://www.cac.mil/Portals/53/Documents/CAC_End_Point_Implementation_Guide_v2.1.1(2010.4.2).pdf)
2. **DoD ID Bar Code Formats — SDK v7.5.0**, DMDC, September 2012. §2.1 / Table 2 (field layout), Appendix B.1 (base-32), Appendix C Tables 25–27 (PDT, Branch, PCC). <https://archive.org/details/DoDIDBarCodeSDKFormatsV750Sep2012>
3. **Appendix C — Sample Barcodes (CAC)**, DMDC. Source of the worked example **1T0QQG0S14PC0MIAA5**. <https://archive.org/details/AppedixCSampleBarcodesCommonAccessCardCAC>
4. **ISO/IEC 16388** — the Code 39 standard. Implemented by your scanner; you do not need to read it.
5. **jkusner/CACBarcode** on GitHub — a Python reference implementation. <https://github.com/jkusner/CACBarcode>